CYLINDER BLOCK

DESCRIPTION	SPECIFICATION	
	Metric	**Standard**
CylinderBore Diameter	98.45 - 98.48 mm	3.8759 - 3.8775 in.
Out of Round (MAX)	0.025 mm	0.001 in.
Taper (MAX)	0.025 mm	0.001 in.
Deck Height	240.03 - 240.18 mm	9.450 - 9.456 in.
Deck Clearance (Below Deck)	0.546 mm	0.0215 in.
Tappet Bore Diameter	23.000 - 23.025 mm	0.9055 - 0.9065
Flatness	0.03 mm per 25 mm 0.05 mm per 152 mm	0.001 in. per 1 in. 0.002 in. per 6 in.
Flatness (MAX)	0.20 mm (MAX) for total length	0.008 in. MAX) for total length
Main Bearing Bore Diameter	68.351 - 68.376 mm	2.691 - 2.692 in.

PISTONS

DESCRIPTION	SPECIFICATION	
	Metric	**Standard**
Weight (Less Pin)	417 - 429 grams	14.7 - 15.1 oz
Piston Pin Bore Diameter	23.650 - 23.658 mm	0.9312 - 0.9315 in.
Piston Pin To Bore (Centerline To Piston Top)	40.61 - 40.72 mm	1.599 - 1.603 in.
Piston To Bore Clearance	0.018 - 0.038 mm	0.0008 - 0.0015 in.
Piston Ring Groove Height		
Compression Rings	1.530 - 1.555 mm	0.0602 - 0.0612 in.
Oil Control Rings	4.035 - 4.060 mm	0.1589 - 0.1598 in.
Piston Ring Groove Diameter		
No. 1 Compression	88.39 - 88.65 mm	3.48 - 3.49 in
No. 2 Compression	87.63 - 87.88 mm	3.45 - 3.46 in.
No. 3 Oil Control	89.66 - 89.92 mm	3.53 - 3.54 in.

PISTON PINS

DESCRIPTION	SPECIFICATION	
	Metric	**Standard**
Piston To Pin Clearance	0.0102 - 0.0208 mm	0.0005 - 0.0009 in.
Diameter	23.637 - 23.640 mm	0.9306 - 0.9307 in.
Piston-Pin To Connecting Rod (Press Fit)	8.9kN	2000 lbf.

PISTON RINGS

DESCRIPTION	SPECIFICATION	
	Metric	**Standard**
Ring Gap		
Top Compression Ring	0.229 - 0.610 mm	0.0090 - 0.0240 in.
Second Compression Ring	0.483 - 0.965 mm	0.0190 - 0.0380 in.
Oil Control (Steel Rails)	0.254 - 1.500 mm	0.010 - 0.060 in.
Side Clearance		
Top Compression Ring	.042 -.084 mm	0.0017 - 0.0033 in.
Second Compression Ring	0.042 - 0.084 mm	0.0017 - 0.0033 in.
Oil Ring (Steel Ring)	.06 -.21 mm	.0024 -.0083 in.